Emergency Hospital Matching System

"Emergency patients are usually taken to the nearest hospital, which often lacks the required treatment or capacity, so critical time is lost travelling to another hospital. We need an Emergency Hospital Matching System that recommends the most suitable hospital (right facilities plus current availability), not just the nearest one. This is my problem statement. Research on possible good solutions." *Generated by LeapSpace*

Emergency Hospital Matching Systems at a Glance

Key Findings Table

Key finding	Population / use case	Why it matters for your system	Evidence
Nearest-hospital routing is often inferior to routing to the nearest appropriate hospital when specialty capability or capacity is mismatched	Stroke, burns, general ED congestion, interfacility transfer models	Supports the core premise of your problem statement: distance alone is not enough	[1] [2] [3] [4] [5] [6]
The strongest real-world evidence is for condition-specific triage-aware routing, especially stroke	Suspected large-vessel-occlusion stroke, trauma, burns, pediatric trauma	Best first target for an MVP is modular routing by emergency category rather than one generic model	[7] [8] [9] [10] [11] [12]
Capability-based routing creates a trade-off: better access to definitive care may come with longer transport or delayed first-line therapy	Mainly stroke; also trauma-system design	Your ranking engine needs explicit detour thresholds and clinician override	[1] [2] [3] [13] [14]
Real-time capacity visibility is feasible, but data quality and interoperability are the main implementation bottlenecks	EMS-hospital coordination, dashboards, bed platforms	Technical success depends as much on governance and workflow fit as on algorithms	[15] [16] [17] [18] [19] [20]
Useful matching requires dynamic capability data, not just total beds	ED, ICU, monitored beds, staffed capacity, specialty coverage	Hospital inputs must include staffed beds, service coverage, diversion/offload status, and time-varying specialty availability	[15] [16] [21] [22] [23] [24]

Key finding	Population / use case	Why it matters for your system	Evidence
ML is most useful as a prediction layer; final destination recommendation should remain rule-based or optimization-based	Dispatch triage, crowding prediction, hospital matching	Hybrid systems are more governable and auditable than pure black-box AI	[25] [26] [27] [28] [29]
Many reported gains are from simulation or retrospective replay; prospective all-condition validation is still limited	EMS rerouting, dispatch optimization, destination support	Build with staged rollout and prospective evaluation, not just offline accuracy claims	[25] [30] [31] [32] [33] [34]

Direct Answer

A good solution to your problem is a **real-time Emergency Hospital Matching System** with three linked layers: **prehospital triage/acuty assessment, live hospital capability/capacity integration, and a transparent ranking engine** that recommends the nearest appropriate hospital rather than the nearest hospital alone [1] [5] [17].

The most evidence-backed design pattern is **specialty-aware destination routing**, especially in stroke, where severity screening linked to thrombectomy-capable centers reduced secondary transfers and improved access to definitive care compared with nearest-center routing, although sometimes with added transport or thrombolysis delay [1] [2] [7] [14].

For a general-purpose system, the recommendation logic should combine:

- **Patient acuity and suspected condition** [35] [36] [37]
- **Required specialty resources** such as trauma, stroke, burn, cardiac, ICU, pediatrics [10] [12] [28]
- **Live hospital status** such as ED crowding, staffed beds, ICU/monitored beds, staff-to-patient ratio, diversion, and specialty coverage [15] [16] [17] [21] [22]
- **Transport time including traffic and handoff delay**, not road distance alone [5] [6] [38]

The literature supports a **hybrid implementation**: use ML to predict high-risk cases and near-term crowding, but use explicit policy rules or optimization for the final recommendation so it remains clinically governable, explainable, and auditable [25] [26] [27] [28].

A practical MVP should start with a few high-impact pathways—**stroke, trauma, STEMI/cardiac, burn, pediatric, and ICU-needing medical emergencies**—plus a ranked destination board showing each hospital's capability match, availability, travel time, and expected delay [7] [12] [17] [33].

Confidence level: High - The core conclusion is strongly supported across multiple domains, though strongest for condition-specific rather than all-condition systems [1] [2] [5] [17].

Study Scope

This report synthesizes evidence from studies and reviews spanning roughly **2015-2025**, with some foundational field-triage work used for context [39] [40]. It covers:

- Prehospital triage and destination routing [1] [10]
- Hospital capacity dashboards and data platforms [15] [41]
- ML and forecasting for triage/crowding [25] [26]
- Optimization models for dispatch, routing, and hospital selection [5] [28]
- Implementation barriers including interoperability, privacy, governance, and workflow [18] [42] [43]

Assumptions & Limitations

- Evidence is strongest for **stroke and specialty pathways**, not for a fully validated all-condition matching platform [7] [12] [29].
- Many destination-control gains are shown in **simulation or retrospective replay** rather than prospective multi-center deployment [25] [30] [34].
- No single retrieved model jointly validates travel time, specialty fit, capacity, equity, and ambulance workload in one operational system [5] [29] [44] [45].
- Real-time capacity data are often incomplete or non-standardized, especially for specialty coverage and staffed capacity [18] [20] [22].

Suggested Further Research

- Prospectively test a **live EMS destination recommender** using acuity + capability + current capacity, with safety and workflow outcomes [31] [33].
- Standardize a **minimum hospital-status dataset** for EMS use: staffed ED beds, ICU beds, monitored beds, diversion, specialty coverage, and offload delay [21] [22] [43].
- Compare **hybrid rule/optimization systems** against pure ML recommenders in real dispatch workflows [27] [46].
- Develop disease-specific **detour thresholds** beyond stroke, especially trauma, burns, and ICU-level medical emergencies [12] [47].
- Evaluate equity and workforce effects of destination matching, not just speed and congestion [29] [44] [45].

Emergency Hospital Matching Systems: Evidence, Design Options, and Implementation Pathways

1. Foundational Context

An Emergency Hospital Matching System is a decision-support system that selects the **best receiving hospital for a patient at that moment**, based on more than geography alone. The evidence base suggests that "best" should be defined as a combination of:

1. **Clinical appropriateness**: the hospital can provide the needed definitive treatment or specialty service [12] [39].
2. **Operational availability**: the required resources are currently usable, not merely listed on paper [15] [17].
3. **Transport feasibility**: travel time and transfer burden remain acceptable for the condition [7] [47].
4. **System impact**: routing should avoid unnecessary crowding, delays, and secondary transfer [5] [6] [23].

This differs from classic nearest-hospital routing, which is still a common baseline policy in emergency systems [23]. The literature repeatedly shows that proximity alone can be misleading when the nearest hospital lacks thrombectomy, trauma, burn, ICU, pediatric, or cath-lab capability, or when it is operationally saturated [1] [4] [10] [24].

Confidence level: High - The conceptual basis is consistent across triage, transfer, and operations studies [5] [17] [23].

2. Why Nearest-Hospital Routing Is Insufficient

Studies across emergency care show several failure modes of nearest-hospital routing.

2.1 Capability mismatch

Stroke studies show that taking suspected large-vessel-occlusion patients to the nearest stroke center rather than a thrombectomy-capable center can increase secondary transfer and delay definitive treatment [1] [3]. Burn studies similarly show that tele-triage and capability-based referral can change transfer decisions substantially, indicating that initial destination choice is often imperfect [4]. Pediatric trauma research reports meaningful undertriage and overtriage even when formal protocols exist, implying that specialty matching remains difficult but important [10].

2.2 Capacity mismatch

Crowding and access block worsen patient flow and are associated with poorer outcomes; real-time destination control using capacity metrics reduced EMS turnaround time and improved balancing across campuses in one study [48] [49]. Models that include expected throughput, case mix, and ED capacity outperform simple proximity logic for reducing overcrowding and time-to-provider [23].

2.3 Secondary transfer burden

A recurring pattern is that sending a patient first to a hospital without the needed service can create large downstream delays. In stroke, secondary transfer from a non-thrombectomy hospital can add around two hours to mechanical thrombectomy access [3]. Delayed burn-center referral was associated with greater resource use and longer stay [50].

Taken together, these findings suggest that your problem statement is well grounded: a system should aim for the **nearest hospital that can actually deliver the needed care now** [1] [23] [51].

Confidence level: High - Supported across multiple specialties and operations literature [1] [4] [48] [49].

3. Triage-Aware Destination Matching

3.1 Severity-based routing for high-acuity emergencies

The best-developed destination-matching evidence is in stroke. Simulation and decision studies found that severity-screened routing to thrombectomy-capable centers can reduce time to groin puncture, improve access to definitive care, and reduce secondary transfers compared with nearest-stroke-center transport [1] [2]. One simulation reported that ROA cutoffs reduced mean time to groin puncture by 29.6 minutes and cut secondary transfers by 60.9% [1]. Guidance-oriented studies describe direct transport to a thrombectomy-capable center when detour is limited, commonly around **15 minutes**, and when it does not preclude thrombolysis [7] [8].

However, the stroke literature also shows the key trade-off: direct routing may improve thrombectomy access but can delay thrombolysis or extend transport, and preliminary trial evidence has not been uniformly supportive [13] [14] [52]. This means a live system should not hard-code one universal bypass policy; it should expose the detour and its consequence [14].

Outside stroke, prehospital ICU-oriented triage and EMS acuity scales show that higher triage levels predict ICU admission and hospital admission, supporting acuity-sensitive destination logic [35]. High-risk identification tools for cardiac ischemia and large-vessel-occlusion stroke also show usable sensitivity/specificity for routing decisions [37] [53].

Design implication

Your system should include a **clinical trigger layer** that classifies the patient into one of a limited number of destination pathways, each with its own capability and detour logic [7] [12].

Confidence level: High - Strong direct evidence for stroke; moderate extension to other high-acuity conditions [1] [2] [35].

3.2 Trauma, burn, and cardiac destination pathways

Trauma systems are the historical foundation for capability-based transport. Field triage criteria were explicitly designed to match injury severity to appropriate trauma-center capability using physiologic, anatomic, mechanism, and special-consideration criteria [39] [40]. But real-world performance is mixed: implementation studies found limited change in actual destination choice in some settings, and older adults remain at risk of undertriage [54] [55] [56] [57]. Pediatric trauma data likewise show substantial undertriage and overtriage [10].

Burn systems offer a practical model for capability matching. Referral criteria and tele-triage pathways reduced unnecessary transfers, improved acuity-based allocation, and reduced provider call burden [58]. Image review changed the initial plan in nearly a quarter of cases, including both down-triage and up-triage, showing that matching improves when capability assessment is richer than simple first-destination assumptions [4].

For cardiac specialty routing, the provided evidence is less conclusive. In out-of-hospital cardiac arrest, bypassing the nearest hospital for specialty-center strategies did not clearly improve short-term survival in one comparative study [59]. This is a useful caution: specialty bypass is not automatically beneficial in every emergency type [59].

Design implication

A general platform should contain **condition-specific routing modules**, not a single one-size-fits-all rule [7] [12] [33].

Confidence level: Medium-High - Strong for the principle, mixed for implementation success outside stroke and burns [10] [11] [58].

3.3 Low-acuity diversion and alternative pathways

Low-acuity diversion is better studied than high-acuity capacity-aware rerouting. Telehealth, treat-in-place, referral, and alternate-destination programs reduced unnecessary ED transport and preserved EMS capacity in selected cohorts [60] [61] [62] [63]. Telephone and on-scene referral pathways appear feasible, though adherence to referral instructions can be imperfect [61]. Dispatch-identified low-acuity groups still include some patients later needing admission or ICU care, so these pathways need safety monitoring [62] [64].

For your problem, low-acuity diversion is not the main use case, but it matters because it can protect ambulance and ED capacity for critically ill patients [36] [60].

Confidence level: Medium - Operationally useful, but not the main evidence base for critical-care destination matching [61] [62].

4. Real-Time Hospital Capacity Integration

4.1 What hospital status signals matter

The literature suggests that matching quality depends on **richer operational signals than bed count alone**. Relevant inputs include:

- Active, staffed, occupied, and available beds [15]
- ICU-bed availability and monitored-bed status [16] [31]
- ED census and crowding metrics [23] [27]
- Staff-to-patient ratios and workforce strain [16] [21]
- Specialty practitioner availability and time-varying service coverage [21] [22]
- Diversion or overload status [31] [33]
- Pending and delayed discharges affecting real usable capacity [17]

This is an important methodological point: a hospital may have physical beds yet lack **staffed beds** or the relevant specialist on call [21] [22].

Confidence level: High - Repeated across dashboard, bed-platform, and operations studies [15] [16] [17].

4.2 Data integration architecture

Several architectures are described for near-real-time aggregation. These combine hospital IT systems, patient administration, EHRs, bed-management systems, dispatch systems, GPS, traffic data, and communication platforms [17] [38] [65]. Cloud and big-data architectures are used for unified storage, aggregation, and rapid transmission; layered collection, buffering, and distributed processing are proposed to reduce processing time [66] [67]. Edge/fog approaches are described as ways to reduce congestion and latency by filtering data before cloud transmission [68].

Some systems have demonstrated operationally useful real-time dashboards, including unit-level bed visibility and national acute-care platforms for ICU-bed search and transfer planning [15] [16] [41].

Practical architecture pattern

A practical matching stack would include:

1. **EMS/dispatch intake**: location, chief complaint, acuity, vitals, crew-entered suspicion [25].
2. **Hospital feed layer**: beds, ICU, monitored beds, staffing, specialty coverage, diversion [16] [17].
3. **Travel-time layer**: GPS and traffic [38].
4. **Prediction layer**: short-term crowding or risk estimates [26] [27].
5. **Ranking engine**: rules/optimization with transparent trade-offs [5] [28].
6. **User interface**: ranked options with reasons and override [33].

Confidence level: High - Feasibility is well supported, though standardization is incomplete [20] [66] [67].

4.3 Interoperability and governance barriers

The most persistent barriers are not algorithmic. Studies repeatedly report incomplete and untimely data, lack of standard formats, privacy concerns, hospital cooperation problems, and workflow mismatch [18] [42] [69] [70]. Proposed solutions include formal data-sharing agreements, common minimum datasets, interoperability programs, reuse of registries and hospital databases, and standards such as structured discrete fields and HL7 FHIR/OMOP-type approaches [20] [42] [43] [71].

A major implementation lesson is that systems must fit real workflows; accurate prediction alone may fail to change operational decisions if it is not embedded into escalation protocols and frontline routines [19] [27] [72].

Confidence level: High - Strong consistency across implementation studies [18] [43] [70].

5. Decision Support and Optimization Approaches

5.1 Rule-based versus learning-based recommendation

ML has shown value in **high-risk recognition**, **admission forecasting**, and **crowding prediction**. One intelligent triage/dispatch system improved hospital matching accuracy from 81.3% to 89.1% and increased high-risk recognition from 82.6% to 90.4% in retrospective replay [25]. Real-time EHR-based admission forecasting outperformed a benchmark rule metric, and some forecasting models were robust enough for operational deployment [26] [73].

But prediction alone is not enough. An explainable overcrowding model predicted well yet did not reliably trigger effective frontline action [27]. This supports a hybrid design: ML estimates future state; policy or optimization converts those estimates into decisions [27] [29].

Recommended stance

Use ML for:

- Acuity/risk estimation [25] [74]
- Near-term crowding/boarding prediction [26] [73]
- Transfer-risk or likely secondary-transfer estimation **(Inferred: low confidence)** based on the documented burden of secondary transfer and mismatch [3] [50]

Use rules/optimization for:

- Hard exclusions and specialty eligibility [7] [39]
- Detour thresholds [7]
- Final hospital ranking [5] [28]

Confidence level: High - The prediction-versus-action distinction is directly supported [25] [27] [29].

5.2 Multi-objective optimization

Recent optimization work increasingly moves beyond shortest travel time. Models separately optimize combinations of response time, wait time, congestion, survival probability, equity, cost, and workload [5] [29] [44] [45]. A 2024 hospital-selection model minimized **total transfer time = travel time + wait time**, reducing waiting times and queues while improving physician-utilization balance [5]. Other models explicitly include hospital specialty in patient allocation alongside travel time and response objectives [28] [75].

However, no single retrieved model fully integrates travel time, specialty match, crowding, equity, and ambulance workload into one validated operational platform [5] [29] [44] [45] [76].

Design implication

For your system, a realistic ranking objective is:

maximize suitability = capability fit + availability fit - travel penalty - congestion penalty - transfer-risk penalty

This formulation is an inference from the optimization and specialty-routing literature rather than a single quoted model [5] [6] [28].
Inferred: low confidence - The components are well supported individually, but the exact composite formula is a synthesis rather than a directly validated universal standard [5] [75].

Confidence level: Medium-High - Strong support for multi-objective logic, but weaker for one universally best formulation [5] [29].

5.3 Congestion-aware routing

A key design lesson is to optimize **total time to effective care**, not just drive time. Models that factor in ED waiting time, receiver-hospital constraints, and congestion show lower queueing and better throughput than proximity-only selection [5] [6] [51]. This directly matches your problem statement: a farther hospital may be faster overall if the nearer one is overloaded or unable to treat [5] [6].

Confidence level: High - Strong consistency across congestion-aware hospital-selection models [5] [6] [77].

6. Validation, Deployment, and Outcomes

Operational results are promising but uneven. Real-time destination control using crowding/capacity metrics reduced EMS turnaround times and improved balancing in one setting [48]. A prospective ED-to-MICU transfer quality-improvement deployment reduced transfer time from 4.2 to 2.2 hours and shortened hospital length of stay [32]. The intelligent triage/dispatch matching study improved matching accuracy and high-risk recognition in retrospective replay [25].

At the same time, live evidence for broad, all-condition EMS hospital matching remains limited. Some systems improved process metrics without clear improvement in longer-term outcomes; for example, one stroke implementation reduced process times but did not improve 90-day functional outcome [78]. CAD-D is described as a destination-recommendation system using patient location, preference, diversion, bed ratios, and chief complaint, but the provided evidence does not include post-deployment outcome metrics [33].

Recommended evaluation metrics

Your system should be judged on:

- Avoided secondary transfers [1] [3]
- Time to definitive treatment [2] [78]
- EMS turnaround and offload delay [22] [48]
- Correct specialty placement / appropriate unit admission [28] [35]
- ED waiting, boarding, and queue metrics [5] [23]
- Safety outcomes such as ICU need, short-term mortality, or deterioration [62] [79]

Confidence level: Medium-High - Good support for process metrics; weaker prospective evidence for hard clinical endpoints in general-purpose systems [25] [32] [34].

7. Design Recommendations for a Practical Emergency Hospital Matching System

7.1 Core inputs and ranking logic

Minimum live inputs should include:

- Patient location and travel time [38]
- Triage acuity and suspected syndrome [35] [53]
- Required specialty resources [12] [28]
- Staffed ED beds, ICU beds, monitored beds [15] [16]
- Diversion or overload status [31] [33]
- Specialty on-call coverage and time-varying service availability [21] [22]
- Expected handoff or waiting delay if available [5] [6]

The system should output a **ranked destination board** with explanation fields such as:

- capability match
- current availability
- travel time
- estimated delay
- detour trade-off
- reason for ranking [25] [33]

7.2 Recommended architecture

A practical architecture is:

1. **Data ingestion layer** from EMS, dispatch, hospital feeds, GPS, traffic [66] [67]
2. **Normalization/interoperability layer** using shared fields/minimum dataset [20] [43]
3. **Prediction layer** for short-horizon crowding and high-risk recognition [25] [26]
4. **Rules/optimization layer** for destination ranking [5] [28]
5. **Clinician override and audit layer** for accountability and learning [69] [80]

7.3 Phased implementation priorities

Phase 1: Silent advisory MVP
Run in parallel without changing destination policy; compare predicted versus actual outcomes [34] [46].

Phase 2: Limited operational use for high-value pathways
Start with stroke, trauma, burn, STEMI/cardiac, pediatric, ICU-needing medical cases [7] [12].

Phase 3: Dynamic load balancing
Add congestion-aware rerouting and short-term capacity prediction once feeds are reliable [5] [26].

Phase 4: Continuous quality improvement
Link destination choice to downstream outcomes using a bidirectional minimum dataset [43] [71].

Confidence level: High - This phased design directly reflects the maturity pattern in the literature: strongest evidence for modular specialty routing plus gradual operationalization [7] [34] [43].

8. Conclusion

The literature supports your problem statement strongly: emergency routing based only on the nearest hospital is often suboptimal because it ignores whether the hospital can deliver the required treatment and whether it can do so **now** [1] [17] [23]. The best-supported solution is not a pure AI black box, but a **hybrid emergency hospital matching platform** that combines triage-aware routing, live capability/capacity data, and transparent optimization or rule logic [5] [25] [27].

The most defensible implementation strategy is to recommend the **nearest appropriate available hospital**, beginning with specialty modules where evidence is strongest, especially stroke and other high-acuity pathways [7] [12]. The largest risks are not mathematical but organizational: weak interoperability, incomplete live data, and poor workflow fit [18] [70]. If those are addressed, the research suggests your proposed system is both justified and technically feasible [6] [66].

Overall confidence level: High - The strategic direction is strongly supported, though evidence for a single fully validated all-condition platform remains limited [1] [5] [17] [34].

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